

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mechanical Engineering) SEMESTER - 6 Winter 2025 (Supply.)

Course :Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTMC602 - MACHINE DESIGN-II

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 The ball bearings are provided with a cage
 - a. to reduce friction
 - b. to facilitate slipping of balls
 - c. to prevent the lubricant from flowing out
 - d. to maintain the balls at a fixed distance apart
 - 2 If a bearing is designated by the number 305, it means that the bearing is of
 - a. light series whose bore is 5 mm
 - b. light series whose bore is 25 mm
 - c. medium series whose bore is 5 mm
 - d. medium series whose bore is 25 mm
 - 3 According to assumptions in the Lewis equation, the gear tooth is treated as _
 - a. a simply supported beam
 - b. a cylinder
 - c. a cone
 - d. a cantilever beam
 - 4 The types of gears used to connect two non-parallel non-intersecting shafts are _____ gears
 - a. Spur
 - b. Helical
 - c. Bevel
 - d. Spiral
 - 5 In designing gear, which of the following is considered for calculating static load?
 - a. Bending stress
 - b. Shear stress
 - c. Fatigue strength
 - d. Compressive stress

- 6 The included angle of a pulley for a V-belt is _____ (in degrees).
 - a. 50-60
 - b. 30-40
 - c. 20-30
 - d. 40-50
- 7 The velocity ratio of two pulleys connected by an open belt or crossed belt is-
 - a. directly proportional to their diameters
 - b. inversely proportional to their diameters
 - c. directly proportional to the square of their diameters
 - d. inversely proportional to the square of their diameters
- 8 The following is an automatic clutch which is controlled by engine speed--
 - a. cone clutch,
 - b. centrifugal clutch ,
 - c. jaw clutch,
 - d. disc clutch
- 9 Single plate clutches are generally used in --
 - a. scooter
 - b. truck
 - c. train
 - d. none of the above
- 10 Which of the following is not a flexible type of connector?
 - a. Belt ,
 - b. Rope,
 - c. chain,
 - d. Gear
- 11 The power transmitted by a belt is maximum when the maximum tension in the belt (T) is equal to _____.
 - a. T
 - b. 2T
 - c. 3T
 - d. 4T
- 12 V-belts are usually used for _____.
 - a. long drives,
 - b. short drives,
 - c. either long drives or short drives,
 - d. neither long drives nor short drives

Q2. Solve the following.

- A) Write Short Notes :- 6
 - 1) Bearing Failure –Causes and remedies
 - 2) Viscosity & Viscosity Index
- B) A single row deep groove ball bearing has a dynamic load carrying capacity of 60,000 N and operates on following work cycle:-Radial load of 4000 N at 400 rpm for 25% of time. Radial load of 6000 N at 500 rpm for 50% of time and, Radial load of 7000 N at 600 rpm for remaining 25 % of time .Calculate the expected life of bearing in hours.

Q3. Solve the following.

- A) Define the Following Terms used in gear drives:- 6
- 1) Circular pitch
 - 2) Backlash
 - 3) Interference & Undercutting
- B) A pair of spur gears with 18 degree full depth involute teeth consists of pinion 6 having 20 teeth meshing with a gear having 40 teeth. The module is 3 and face width is 36 mm. The material of pinion and gear is steel with a ultimate tensile strength of 450 N/mm^2 . The gears are heat treated to surface hardness of 350 BHN. The pinion rotates at 1400 rpm and service factor for the application is 1.2. Assume the velocity factor accounts for the dynamic load and factor of safety is 2. Determine the power that the gear can transmit. Assume Lewis form factor $Y = 0.3$.

Q4. Solve Any Two of the following.

- A) Explain with help of neat diagram classification & types of Gears. 6
- B) A pair of parallel helical gears transmitting 10 KW power consists of 24 teeth 6 pinion running at 3000 rpm. The speed ratio is 3. The normal pressure angle is 20 degree and helix angle is 43 degree. Both the gears are made of same material having ultimate strength of 400 N/mm^2 . If the service factor is 1.5 and factor of safety is 2, Calculate:-
- 1) Normal Module, considering velocity factor accounts for the dynamic load and that the face width is ten times the normal module & the pitch line velocity is 10 m/s.
 - 2) Consider the first preference value of normal module and find out main dimensions of the gears. Assume form factor $Y = 0.438$
- C) A pair of bevel gears having 20 pressure angle consists of a pinion having 18 6 teeth meshing with a gear having 36 teeth. The face width is 22 mm and module is 3. The material of gear and pinion is steel 50C4 ($S_{ut} = 700 \text{ N/mm}^2$) The gear teeth are hardened, ground and lapped with hardness of 450 BHN. The pinion rotates at 400 rpm and receives 4 KW power from electric motor. The starting torque is 150 % of the rated torque. Find out the effective load acting on the gears. Use following values if required $C = 11400 \text{ N/mm}^2$, $e = 0.0125$.

Q5. Solve Any Two of the following.

- A) Differentiate between Bevel Gears and Worm Gears. 6
- B) A pair of worm and worm wheel is designated as 3/60/10/6 the worm is 6 transmitting 5 KW at 1200 rpm to the worm wheel. The coefficient of friction is 0.2 and normal pressure angle is 18 degree. Determine the components of the gear tooth force acting on the worm and worm wheel.
- C) Design a flat belt drive to operate dynamo generating 20 KW at 1800 rpm. The 6 pulley mounted on it is having diameter of 180 mm.
- Use the following data:-
- 1) Allowable stress for the belt = 3 MPa.
 - 2) Thickness of the belt = 12 mm
 - 3) Density of rubber belt = 1000 kg/m^3
 - 4) Angle of contact at dynamo pulley = 160 Degree.
 - 5) Coefficient of friction between belt and pulley = 0.2
 - 6) Dynamo efficiency = 80%

Q6. Solve Any Two of the following.

- A) Differentiate between belt drives and chain drives. 6

- B) A centrifugal clutch transmits 20 KW at 800 rpm consists of 4 shoes. The clutch is to be engaged at 500 rpm. The inner radius of drum is 160 mm. The center of gravity of shoe is 140 mm when it is engaged. The coefficient of friction is 0.3, while the intensity of pressure is 0.1 N/mm^2 , Calculate
- 1) mass of each shoe
 - 2) Dimensions of friction lining.
- C) An electric motor 100 KW running at 800 rpm runs a compressor at 300 rpm using 6 V belt drive. The diameter of the pulley mounted on compressor shaft is 1 meter and the center distance between two pulleys is 2 meter. The belt speed is limited to 1400 m/min. Determine the number of V belts & the length of each belt required to transmit the power, if each belt has cross sectional area of 400 mm^2 , density 1000 kg/m^3 and an allowable tensile stress of 3 MPa. The groove angle of the pulley is 36 degree & the coefficient of friction between belt and pulley is 0.25.

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